$$= \frac{1}{5}\begin{bmatrix} 0-280+380 \\ 960+0-760 \\ -640+560+380 \end{bmatrix}$$

$$= \frac{1}{5}\begin{bmatrix} 100 \\ 200 \\ 300 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 20 \\ 40 \\ 60 \end{bmatrix}$$

$x = 20$, $y = 40$ and $z = 60$.

Cost of 1 kg onion is ₹20, cost of 1 kg wheat is ₹40 and cost of 1 kg rice is ₹60.

Exercise 1.3

1. Solve by matrix inversion method:
   $2x + 3y - 5 = 0$; $x - 2y + 1 = 0$
2. Solve by matrix inversion method:
   (i) $3x - y + 2z = 13$; $2x + y - z = 3$; $x + 3y - 5z = -8$
   (ii) $x - y + 2z = 3$; $2x + z = 1$; $3x + 2y + z = 4$.
   (iii) $2x - z = 0$; $5x + y = 4$; $y + 3z = 5$
3. A sales person Ravi has the following record of sales for the month of January, February and March 2009 for three products A, B and C He has been paid a commission at fixed rate per unit but at varying rates for products A, B and C

Months	Sales in Units			Commission
	A	*B*	*C*	
January	9	10	2	800
February	15	5	4	900
March	6	10	3	850

   Find the rate of commission payable on A, B and C per unit sold using matrix inversion method.
4. The prices of three commodities A, B and C are ₹x, ₹y and ₹z per unit respectively. P purchases 4 units of C and sells 3 units of A and 5 units of B. Q purchases 3 units of B and sells 2 units of A and 1 unit of C. R purchases 1 unit of A and sells 4 units of B and 6 units of C. In the process P, Q and R earn ₹6,000, ₹5,000 and ₹13,000 respectively. By using matrix inversion method, find the prices per unit of A, B and C.
5. The sum of three numbers is 20. If we multiply the first by 2 and add the second number and subtract the third we get 23. If we multiply the first by 3 and add second and third to it, we get 46. By using matrix inversion method find the numbers.
6. Weekly expenditure in an office for three weeks is given as follows. Assuming that the salary in all the three weeks of different categories of staff did not vary, calculate the salary for each type of staff, using matrix inversion method.

Week	Number of employees			Total weekly salary (in ₹)
	A	B	C	
1st week	4	2	3	4900
2nd week	3	3	2	4500
3rd week	4	3	4	5800

1.3 Input–Output Analysis

Input–Output analysis is a technique which was invented by Prof. Wassily W. Leontief. Input–Output analysis is a form of economic analysis based on the interdependencies between economic

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